FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C
Sol. $10^{-3}(1 + R) = 10$
$\therefore R = 9999\Omega$

2. C
Sol. Factual

3. B
Sol. $\Delta V = \frac{B\omega^2(\ell)}{2} = \frac{B\omega^2(\sqrt{3}\,R)}{2} = V_B - V_A.$

4. A
Sol. $W = \int Fvdt$

5. C
Sol: Since, D_1 will be acting as short circuit, D_2 will act as open circuit so,
Current through 5Ω resistance and diode D_1 is $\frac{20}{5} = 4$ A

6. D
Sol. Factual

7. B
Sol: $\cos(180 - \theta) = A/B \Rightarrow \theta = \cos^{-1}\left(-\frac{A}{B}\right)$

8. B

Sol. At steady state.

Rate of generation of A = Rate of decay of A.

$q = \lambda N_A$

$\Rightarrow \quad N_A = \dfrac{q}{\lambda}$

9. A

Sol. $\dfrac{1}{2}mu^2 = 2mg \sin\theta \, \ell$

10. D

Sol. F is same for all.

Use $F = I(\vec{\ell} \times \vec{B})$

11. A

Sol. P must be applied at centre of mass.

$\Rightarrow \quad \dfrac{4\ell}{3}$

12. A

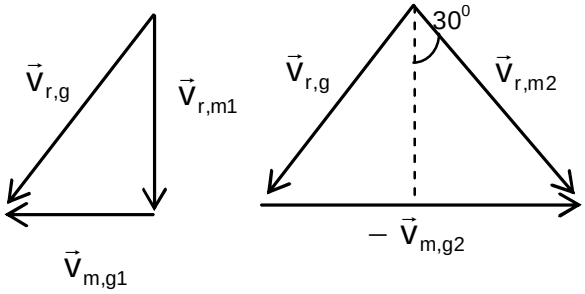
Sol. Time period is independent of mass and depends only upon the distance from the planet.

13. B

Sol. If ΔV is increased λ will decrease and hence fringe width will decrease.

14. D

Sol.



15. C

Sol: Velocity of pendulum bob in mean position

$v_1 = \sqrt{2gh} = \sqrt{2\times \ 10\times 5\times 10^{-2}} \ = 1$ m/s

When the bob picks up a mass 10^{-3} kg at the bottom, then by conservation of linear momentum the velocity of coalesced mass is given by

$m_1v_1 + m_2v_2 = (m_1 + m_2)v$

$10^{-2} + 10^3\times 0 = (10^{-2}+10^{-3})\ v$

or $v \ = \ \dfrac{10^{-2}}{1.1 \ \times 10^{-2}} \ = \ \dfrac{10}{11} \ \ m/s$

if h′ is the height risen by combined mass, we have

$$\frac{1}{2}\left(m_1+m_2\right)v^2 = \left(m_1+m_2\right) gh$$

$$\therefore h' = \frac{v^2}{2g} = \frac{\left(\frac{10}{11}\right)^2}{2 \times 10} m = \frac{5}{121} m = 0.413m$$

16. A

Sol. In S.H.M. displacement

$y = a\sin(\omega t + \phi)$; velocity, $v = a\omega\cos(\omega t + \phi)$

Let $t = t'$ be the instant, when $y = y' = 25$cm

And $v = v' = 100$ cm/s.

Given, $\omega = 4$ rad/s

Then, $25 = a\sin(\omega t' + \phi)$ and $100 = a\omega\cos(\omega t' + \phi)$

Or $25 = a\sin(4t' + \phi)$ and $100 = a4\cos(4t' + \phi)$

Squaring and adding we get

$625 + 625 = a^2$ or $a = 25\sqrt{2}$ cm

Dividing, we get;

$\tan(4t' + \phi) = 1$ or $4t' + \phi = \frac{\pi}{4}$

After time 2.4sec from the instant t′, the time from the beginning, $t = t' + 2.4$, we have

$y = 25\sqrt{2}\sin[4(t' + 2.4) + \phi]$

$= 25\sqrt{2}\sin[4t' + 9.6 + \phi]$

$=25\sqrt{2}\sin\left[\frac{\pi}{4} + 9.6\right] = -28.68$ cm

and $v = 25\sqrt{2} \times 4\cos[4(t' + 2.4) + \phi]$

$= 100\sqrt{2} \times 4\cos\left[\frac{\pi}{4} + 9.6\right] = -82.18$ cm/s.

17. B

Sol. As the velocity increases pressure decreases hence the height decreases.

18. B

Sol. Using path difference.

19. C

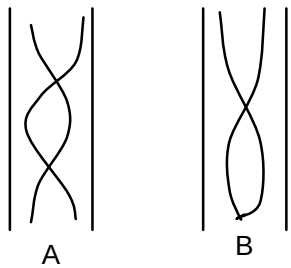
Sol. (a) $2\frac{V_A}{2L_A} = 3\frac{V_B}{4L_B}$

$$\Rightarrow \frac{V_A}{V_B} = \frac{3}{4} \qquad \left[\because L_A = L_B\right]$$

$$V_A = \sqrt{\frac{\gamma_A RT}{M_A}}, \qquad V_B = \sqrt{\frac{\gamma_B RT}{M_B}}$$

$$\Rightarrow \sqrt{\frac{\gamma_A}{\gamma_B}\frac{M_B}{M_A}} = \frac{3}{4},$$

$$\gamma_A = \frac{5}{3}, \qquad \gamma_B = \frac{7}{5}$$



$$\Rightarrow \frac{M_A}{M_B} = \frac{400}{189}$$

20. A

Sol. $\omega = \dfrac{v}{\ell \sin\theta}$

$V_c = \dfrac{v}{2\sin\theta}$

SECTION – B

21. 0

Sol. $V_A - V_B = i(5\Omega) + 10V + L\dfrac{di}{dt}$

$= 5(5) + 10 + L(-10^{+3})$

$V_A - V_B = 35 - 35 \times 10^{-3} \times 10^{+3} = 0.$

22. 3

Sol. V = 0.6 m

$\Rightarrow \quad \dfrac{h_I}{h_o} = \dfrac{h_I}{0.05} = 2 \Rightarrow h_I = 0.1$ cm.

$\Rightarrow$ Distance between them = 0.3 cm = 3 mm.

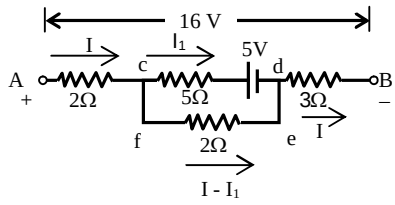
SECTION – C

23. 00002.40

Sol. Required distance = $2 \times \dfrac{D\lambda}{d} = \dfrac{2\times 2\times 600\times 10^{-9}}{1\times 10^{-3}}$ m = 2.40 mm.

24. 00000.06

Sol. $\phi_A - \phi_B = 16$ …(i)



Apply KVL on loop (cd efc)

$0 - 5I_1 - 5 + 2(I - I_1) = 0$

$\Rightarrow I_1 = \dfrac{1}{7}(2I - 5)$ …(ii)

Apply KVL on path (AC d B)

$\phi_A - 2I - 5I_1 - 5 - 3I = \phi_B$

$\Rightarrow \phi_A - \phi_B = 5I + 5I_1 + 5$

$\Rightarrow 16 = 5I + 5\left(\dfrac{2I-5}{7}\right) + 5$

$\Rightarrow 16 - 5 + \dfrac{25}{7} = I\left(5 + \dfrac{10}{7}\right)$

$\Rightarrow I = \dfrac{34}{15}$ Amp

(i) and (ii) give

$I_1 = -\frac{1}{15}$ Amp

Current through battery (5V) = $\frac{1}{15}$ Amp = 0.06 Amp

25. 00066.66

Sol. For series combination $R = 220^2\left(\frac{1}{200}+\frac{1}{100}\right) = 220^2 \times \frac{3}{200}$

$\therefore$ Power $= \frac{220^2}{R} = \frac{200}{3}$ W = 66.66

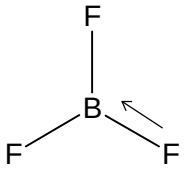
Chemistry

PART – II

SECTION – A

26. B

Sol. Due to existence of $p\pi - p\pi$ back bonding in BF_3.

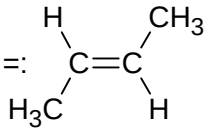


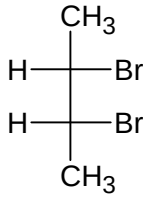
27. B

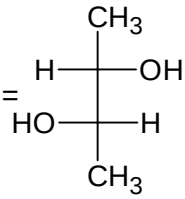
Sol. $KO_2 \xrightarrow{H_2O} KOH + H_2O_2 + O_2$

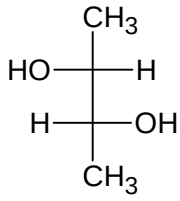
$K_2S \xrightarrow{H_2O} KOH + H_2S$

28. D

Sol. P =: 

Q = 

R or S = 

or 

29. C

Sol. Due to double bond, the geometrical isomers are:

$$CH_3CH = CH - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH = CH - CH_3$$

Cis	Cis
Trans	Trans
Cis	Trans
Trans	Cis

} Identical

When the double bonds are Cis and trans, the molecule becomes optically active.

$\therefore$ For trans Cis(two optical isomers are possible)

$\therefore$ Total no. of stereoisomers are:

Geometrical(2) + Optical(2) = 4

30. C

Sol. $2\pi r = n\lambda \Rightarrow \lambda = \frac{2\pi r}{n} = \frac{2\pi n^2 r_0}{n \times z} = 2\pi n r_0 = 2\pi \times 4 \times r_0 = 8\pi r_0$

31. A

Sol. C represents most probable velocity.

32. C

Sol. Catalyst only fastens the reaction.

33. B

Sol. $CO_3^{2-} + H^+ \longrightarrow HCO_3^-$ (Weak base does not dissociate much). Therefore, the reaction proceeds to forward direction by removing CO_3^{2-} as HCO_3^-.

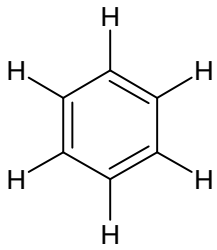
34. D

Sol. Isochoric process have $\Delta V = 0$, so work $W = P\Delta V = 0$

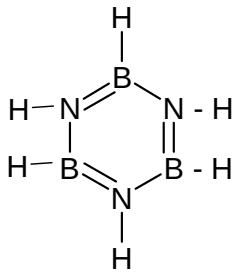
35. D

Sol.

Organic benzene



Inorganic benzene



36. D

Sol. The nearer the –I groups towards COOH group, stronger is the acid.

37. B

Sol. $2NaHSO_4 \xrightarrow{\Delta} Na_2SO_4 + SO_3 + H_2O$

38. C

Sol. $Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$

39. D

Sol. (D) contains maximum number of non-identical 'H' atoms which are to be substituted by chlorine.

40. B

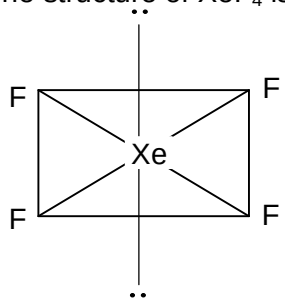
Sol. A, C, D will show geometrical isomerism. (B) will show geometrical as well as optical isomerism.

41. B

Sol. Aniline will be deactiviated in presence of H^+.

42. C

Sol. The structure of XeF_4 is:



43. C

Sol. For zero order reaction,
Rate = K
Both rate and rate constant are independent of concentration.

44. D

Sol. pH of $NaH_2PO_4 = \frac{p^{K_{a_1}} + p^{K_{a_2}}}{2} = \frac{7.2 + 10.6}{2} = 8.9$

45. C

Sol. In NH_4Cl, the oxidation state of nitrogen is -3.

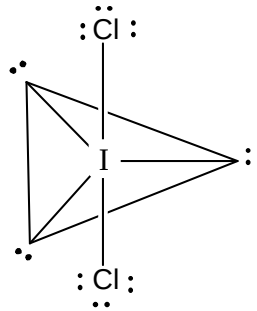
SECTION – B

46. 4

Sol. O_2, O_2^+, H_2^+ and NO are paramagnetic

47. 9

Sol. Structure of ICl_2^-



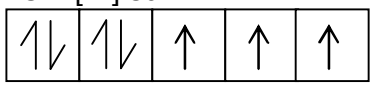
SECTION – C

48. 00003.87

Sol. $H^+ = \sqrt{K_a C} = \sqrt{1.8 \times 10^{-5} \times 10^{-3}} = 1.34 \times 10^{-4}$
pH = -log[H^+] = 3.87

49. 3.87

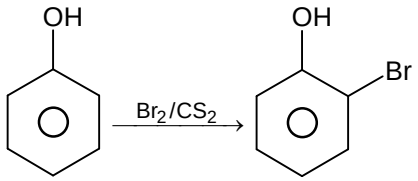
Sol. EC = [Ar] $3d^7$



$\mu spin = \sqrt{n(n+2)} = \sqrt{3 \times 5} = 3.87$

50. 00043.25

Sol.



$M_o = 173$

So, $\frac{M_o}{4} = 43.25$

Mathematics

PART – III

SECTION – A

51. A

Sol. $y = 2\log_a x$

$\Rightarrow \log_a x = \frac{y}{2}$

$\Rightarrow x = a^{y/2}$

$\Rightarrow f^{-1}(y) = a^{y/2}$

$\Rightarrow f^{-1}(b+c) = a^{\frac{b+c}{2}} = f^{-1}(b).f^{-1}(c)$

52. B

Sol. $f(x) = \frac{\ln(\pi + x)}{\ln(e + x)}$

$$f'(x) = \frac{\frac{\ln(e+x)}{(\pi+x)} - \frac{\ln(\pi+x)}{(e+x)}}{(\ln(e+x))^2}$$

Now, consider the function $g(y) = y\ln y$

$g'(y) = \ln y + 1$

$g'(y) > 0$ when $y > \frac{1}{e}$

$g(y)$ is an increasing function when $x > \frac{1}{e}$

$\Rightarrow (e+x)\ln(e+x) < (\pi+x)\ln(\pi+x) \qquad (x \geq 0)$

$\Rightarrow \frac{\ln(e+x)}{(\pi+x)} - \frac{\ln(\pi+x)}{(e+x)} < 0$

$\Rightarrow f'(x) < 0 \ \forall x \geq 0$

$\Rightarrow f(x)$ is decreasing is $[0, \infty)$

53. A

Sol. $x^3 + y^3 + (-1)^3 - 3(x)(y)(-1) = 0$

$\Rightarrow (x + y - 1) = 0$ OR $x^2 + y^2 - xy + x + y + 1 = 0$

54. D

Sol. From the given equation we get

$S_1 = \tan\theta_1 + \tan\theta_2 + \tan\theta_3 + \tan\theta_4 = \sin 2\beta$.

$S_2 = \sum \tan\theta_1 \tan\theta_2 = \cos 2\beta$

$S_3 = \sum \tan\theta_1 \tan\theta_2 \tan\theta_3 = \cos\beta$

and $S_4 = \tan\theta_1 \tan\theta_2 \tan\theta_3 \tan\theta_4 = -\sin\beta$

Now $\tan(\theta_1+\theta_2+\theta_3+\theta_4) = \dfrac{S_1 - S_3}{1 - S_2 + S_4}$.

$= \dfrac{\sin 2\beta - \cos\beta}{1 - \cos 2\beta - \sin\beta} = \dfrac{\cos\beta(2\sin\beta - 1)}{\sin\beta(2\sin\beta - 1)} = \cot\beta$

55. A

Sol. $\arg\left(\dfrac{z-2i}{z+2i}\right) = \dfrac{\pi}{6}$

$\Rightarrow \arg(z-2i) - \arg(z+2i) = \dfrac{\pi}{6}$

$\Rightarrow \tan^{-1}\left(\dfrac{y-2}{x}\right) - \tan^{-1}\left(\dfrac{y+2}{x}\right) = \dfrac{\pi}{6}$

$\Rightarrow \dfrac{xy - 2x - xy - 2x}{x^2 + y^2 - 4} = \dfrac{1}{\sqrt{3}}$

$\Rightarrow x^2 + y^2 - 4 = -4\sqrt{3}\,x$

$\Rightarrow x^2 + y^2 + 4\sqrt{3}\,x - 4 = 0$

$\Rightarrow \left(x + 2\sqrt{3}\right)^2 + y^2 = 12 + 4 = 4^2$

$\therefore$ centre $\left(-2\sqrt{3}, 0\right)$, radius = 4

56. B

Sol. $I = \int \dfrac{\sin 3x}{32}\,dx$

57. C

Sol. $\dfrac{f''(x)}{f'(x)} = 1$

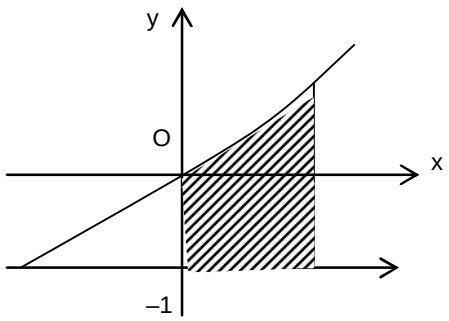
Integrating $\quad \ln f'(x) = x + c,\ f'(0) = 1 \Rightarrow c = 0$

$f'(x) = e^x$

$f(x) = e^x + k,\ f(0) = 0 \Rightarrow k = -1$

$f(x) = e^x - 1$

Area $= \displaystyle\int_0^1 (e^x - 1 + 1)\,dx = e^x\Big]_0^1 = e - 1$



58. A

Sol. $\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1) \geq -\dfrac{3}{2}$

59. A

Sol. $\frac{dy}{dx}=\frac{4}{2}e^{\frac{1-x}{2}}+1+x+x^2=2e^{\left(\frac{1-x}{2}\right)}+x^2+x+1$

$g'(y)=\frac{dx}{dy}=\frac{1}{2e^{-\left(\frac{x-1}{2}\right)}+x^2+x+1}$

When $y=-\frac{7}{6}$ then $x=1$

$\left.\frac{dy}{dx}\right]_{x=-7/6}=\frac{1}{2+3}=\frac{1}{5}$

60. A

Sol. $\frac{\left(\alpha^5+\beta^5\right)}{\left(\alpha^3+\beta^3\right)}=\frac{11}{3}$

$\frac{1}{\left(\alpha^2+\beta^2-\alpha\beta\right)}\left(\alpha^4+\beta^4-\alpha\beta\left(\alpha^2+\beta^2-\alpha\beta\right)\right)=\frac{11}{3}$

$\frac{\left(\alpha^4+\beta^4\right)}{\left(\alpha^2+\beta^2-\alpha\beta\right)}-\alpha\beta=\frac{11}{3}$

$\frac{25-2(\alpha\beta)^2}{5-\alpha\beta}-\alpha\beta=\frac{11}{3}$

Let $\alpha\beta=t$; by coefficient $\alpha\beta=2$

61. D

Sol. Number of ways = (Number of one-one function from set A = $\{s_1, s_2 \ldots s_6\}$ to A such that $f(s_i)\neq s_i\times 6!=D_6\times 6!$

62. C

Sol. The given plane passes through $\vec{a}$ and is parallel to the vector $\vec{b}-\vec{a}$ and $\vec{c}$. So it is normal to $\left(\vec{b}-\vec{a}\right)\times\vec{c}$. Hence, its equation is $\left(\vec{r}-\vec{a}\right).\left(\left(\vec{b}-\vec{a}\right)\times\vec{c}\right)=0$ or

$\vec{r}.\left(\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right)=\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$

The length of the perpendicular from the origin to this plane is $\frac{\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]}{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|}$

63. A

Sol. $A^n=\begin{bmatrix}\cos n\theta & \sin n\theta\\ -\sin n\theta & \cos n\theta\end{bmatrix}$

$\therefore \lim_{n\to\infty}\frac{A^n}{n}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}$

64. A

Sol. D.R's of PQ are x_1-1, y_1-1, z_1-1

$\Rightarrow 1(x_1-1)+1(y_1-1)+1(z_1-1)=0$

$\Rightarrow x_1+y_1+z_1=3$

Also $x_1+2y_1+3z_1=4$

$x_1+y_1+z_1=3$

$\Rightarrow (x_1,y_1,z_1)$ is a point on the two planes

$x+2y+3z=4$

$x+y+z=3$

$\therefore$ line of intersection of these two planes will be the required locus which is

$\frac{x}{1}=\frac{y-5}{-2}=\frac{z+2}{1}$

65. B

Sol. Parabola is of the form $(x+3)^2+\left(y-\frac{1}{2}\right)^2=\left(\frac{3x+4y-\frac{1}{2}}{5}\right)^2$ having focus $\left(-3,\frac{1}{2}\right)$ and point of intersection of lines given by the equation $2x^2-3y^2+xy-3x-7y+2=0$ can be obtained by solving equations $4x+y-3=0$ and $-6x+x-7=0$ (partial differentiation method), i.e $(1-1)\equiv$ centre (point of intersection of normals).

Hence required circle is $(x-1)^2+(y+1)^2=(1+3)^3+\left(-1-\frac{1}{2}\right)^2$

Or $x^2+y^2-2x+2y+2=16+\frac{9}{4}$ i.e. $4x^2+4y^2-8x+8y-65=0$

66. D

Sol. Asymptotes equations are $x^2-xy+k=0$

67. C

Sol. $\frac{dy}{dx}=\frac{y}{x}\left(\ell n\frac{y}{x}-1\right)$

Put $\frac{y}{x}=t$

68. A

Sol. $T_1=\frac{1}{4}(4-3)=1-\frac{3}{4}$

$T_2=\frac{1(3)}{4(6)}[6-5]=\frac{3}{4}-\frac{1(3)(5)}{4(6)}$

$T_3=\frac{1(3)(5)}{4(6)(8)}[8-7]=\frac{1(3)(5)}{4(6)}-\frac{1(3)(5)(7)}{4(6)(8)}$

$\vdots$

$\therefore T_1+T_2+T_3.....=1$

69. C

Sol. The expression can be interpreted as the square of distance between the points $(\tan A, \cot A)$ and $(\cos B, \sin B)$

The minimum value of this distance is the minimum distance between the curves $xy = 1$ and $x^2 + y^2 = 1$

70. A

Sol. Any element, is of the form of $\frac{8!}{t_1!t_2!t_3!t_4!}\left(x^2\right)^{t_1}\left(\frac{1}{x^2}\right)^{t_2}(y)^{t_3}\left(\frac{1}{y}\right)^{t_4}$ where

$t_1 + t_2 + t_3 + t_4 = 8;\ t_1 \geq 0$

The constant term occur when $t_1 = t_2$ and $t_3 = t_4$

So, $t_1 + t_3 = 4 \Rightarrow (0,4),(4,0);(1,3),(3,1);(2,2)$

$\Rightarrow$ Constant term:

$$\frac{8!}{0!0!4!4!} + \frac{8!}{4!4!0!0!} + \frac{8!}{1!1!3!3!} + \frac{8!}{3!3!1!1!} + \frac{8!}{2!2!2!2!}$$

SECTION – B

71. 2

Sol. When $\cot x \geq 0$

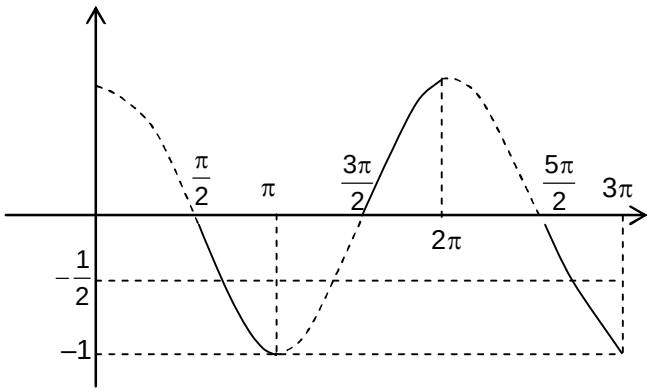
$\Rightarrow \frac{1}{\sin x} = 0$

$\Rightarrow$ No possible value of x.

When $\cot x < 0 \quad \Leftrightarrow \quad x \in \left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{3\pi}{2}, 2\pi\right) \cup \left(\frac{5\pi}{2}, 3\pi\right)$

$\Rightarrow -\cot x = \cot x + \frac{1}{\sin x}$

$\Rightarrow \cos x = \frac{-1}{2}$



Number of solution = 2

72. 1

Sol. Point of intersection of curves is (1, 0)

$y = 3^{x-1}\ln x$

$\Rightarrow y' = \dfrac{3^{x-1}}{x} + \ln x.3^{x-1}.\ln 3$

$\therefore m_1 = 1$

Now, $y = x^x(1+\ln x)$

$\therefore\ m_2 = 1$

So, $\tan\theta = \tan\theta = \left|\dfrac{1-1}{1+1}\right| = 0$

$\Rightarrow \theta = 0 \Rightarrow \cos 0^o = 1$

SECTION – C

73. 00002.75

Sol. We have $T = \int_0^{\ln 2} \dfrac{\left(3e^{3x} + 2e^{2x} - e^x\right) - \left(e^{3x} + e^{2x} - e^x + 1\right)}{e^{3x} + e^{2x} - e^x + 1} dx$

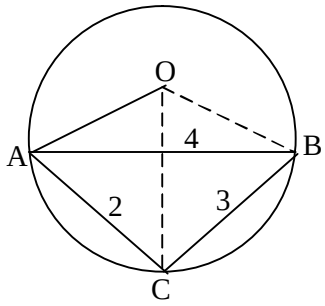
$= \left[\ln\left(e^{3x} + e^{2x} - e^x + 1\right) - x\right]_0^{\ln 2}$

$= \ln\dfrac{11}{2} - \ln 2 = \ln\dfrac{11}{4}$

$\Rightarrow e^T = e^{\ln\frac{11}{4}} = \dfrac{11}{4}$

74. 00531.25

Sol. Using the property that equal chords subtends equal angles at centre of circle, then problem can be converted to the diagram in adjoining figure



$AB = 4, AC = 2, BC = 3$

$\angle ABC = \alpha/2$

$\cos(\alpha/2) = \dfrac{9+16-4}{2\times 3\times 4} = \dfrac{7}{8} \Rightarrow \cos\alpha = 2\cos^2(\alpha/2) - 1$

$= 2\times\dfrac{49}{64} - 1 = \dfrac{98-64}{64} \Rightarrow \cos\alpha = \dfrac{34}{64} = \dfrac{17}{32}$

75. 00000.25

Sol. For every selection of 6 numbers without loss of generality there will be 5 favorable cases out of every 20 cases.